

Regularized MAR and LSTM in forecasting IV surfaces

Trading and Big Money

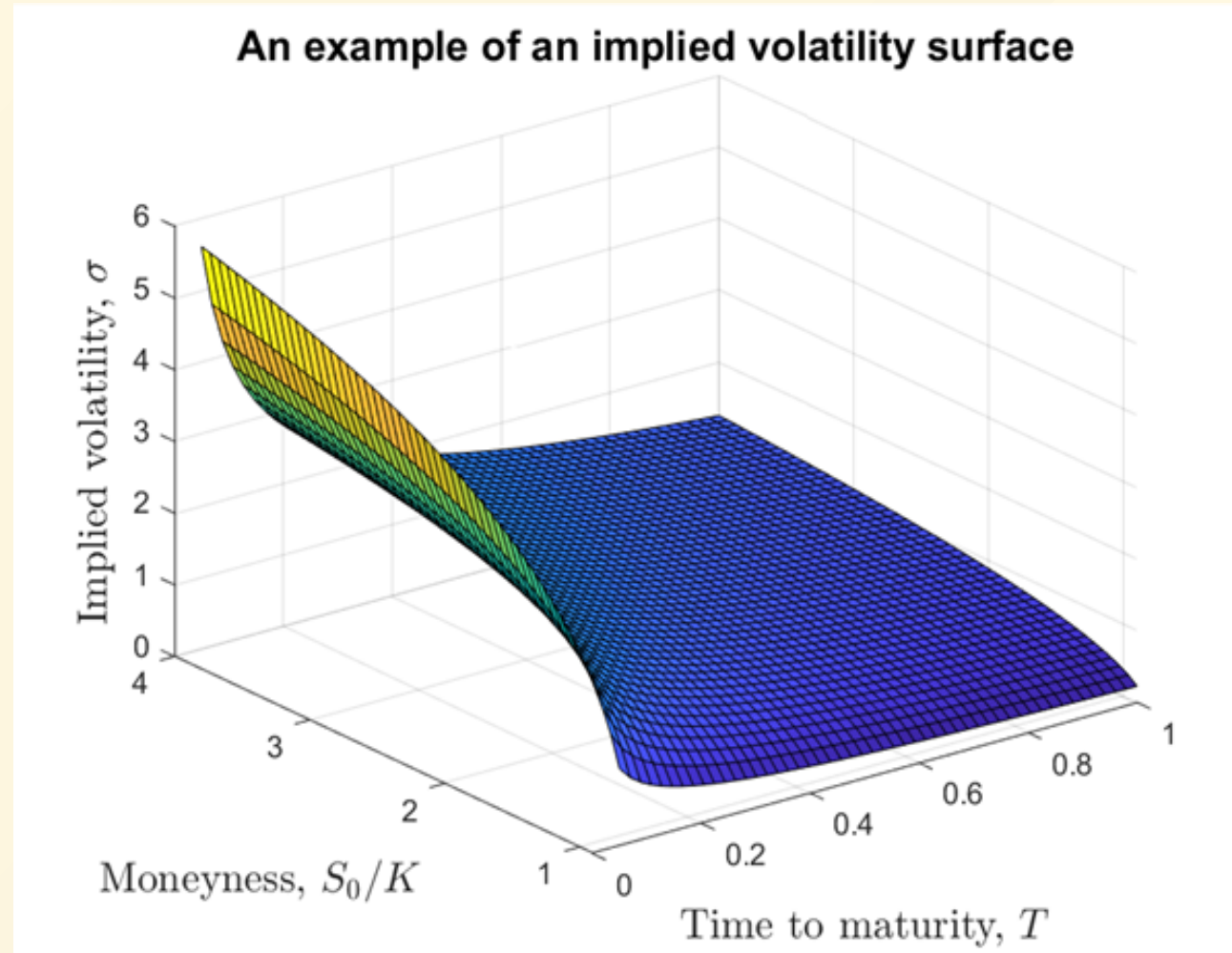
Company	Headcount	Compensation
Jane Street	3K	Mean 2.68M
Google	190K	Median 331K

Source: Yahoo Finance

Option Pricing Over(simplified)view

- **Equity option:** equity derivative that gives the owner the right, but not the obligation, to execute a trade.
- **Two types:** call & put options
 - Strike price
 - Expiration day
 - Premium
- **Implied Volatility (IV):** obtained by inverting the Black-Scholes model (*Black & Scholes, 1973*)

Implied Volatility (IV) Surface



IV Estimation Issues

- **Deep learning:** LSTM (Hochreiter, Schmidhuber, 1997) - highly accurate, but...
 - considered as "black-box"
 - **MODEL RISK = FINANCIAL LOSS**
- **Econometrics:** VAR (Sims, 1980) - highly interpretable, but...
 - fail to preserve matrix structure
 - have large # of coefficients
- **Fortunately, MAR Lasso** (*Chen et al., 2020*) has the best of both worlds

MAR Lasso vs. LSTM: Lack of Literature

- No comparison:
 - Regularized MAR (Jiang, Shen, Li, Gao, 2024)
 - MAR extension - SIGMAR (Wu, Yang, Yan, Feng, 2025)
- No MAR Lasso:
 - ConvLSTM/LSTM/VAR/VEC (Medvedev, 2019)
 - LSTM/GARCH/GJRGARCH (Chatterjee, Bhowmick, Sen, 2022)
- No IV:
 - LSTM/MAR/VAR (Liu, Fang, 2026)

Purpose

- Compare LSTM and MAR Lasso in Forecasting IV Surfaces
- Research questions:
 1. Does MAR Lasso outperform LSTM in predicting IV surfaces for a 1-day, 30-day, and 90-day horizon?
 2. Does MAR Lasso significantly improve upon unconstrained MAR and VAR in predicting IV surfaces?

Methodology (1): Why not VAR? Too bad!

VAR: Vector Autoregressive Matrix

$$\text{vec}(X_t) = \Phi \text{vec}(X_{t-1}) + \text{vec}(E_t)$$

where $X_t \in \mathbb{R}^{m \times n}$ represents the IV surface at time t , and $\text{vec}(\cdot)$ is the vectorization of a matrix by stacking its columns.

- Fail to capture relationship between rows and columns
- Number of parameters: $O(m^2 n^2)$

Methodology (2): MAR? Better!

MAR: Matrix Autoregressive Model

$$X_t = AX_{t-1}B^\top + E_t$$

where $\otimes$ denotes the Kronecker product.

- Number of parameters: $O(m^2 + n^2)$

Methodology (3): MAR Lasso? Much Better! Maybe...

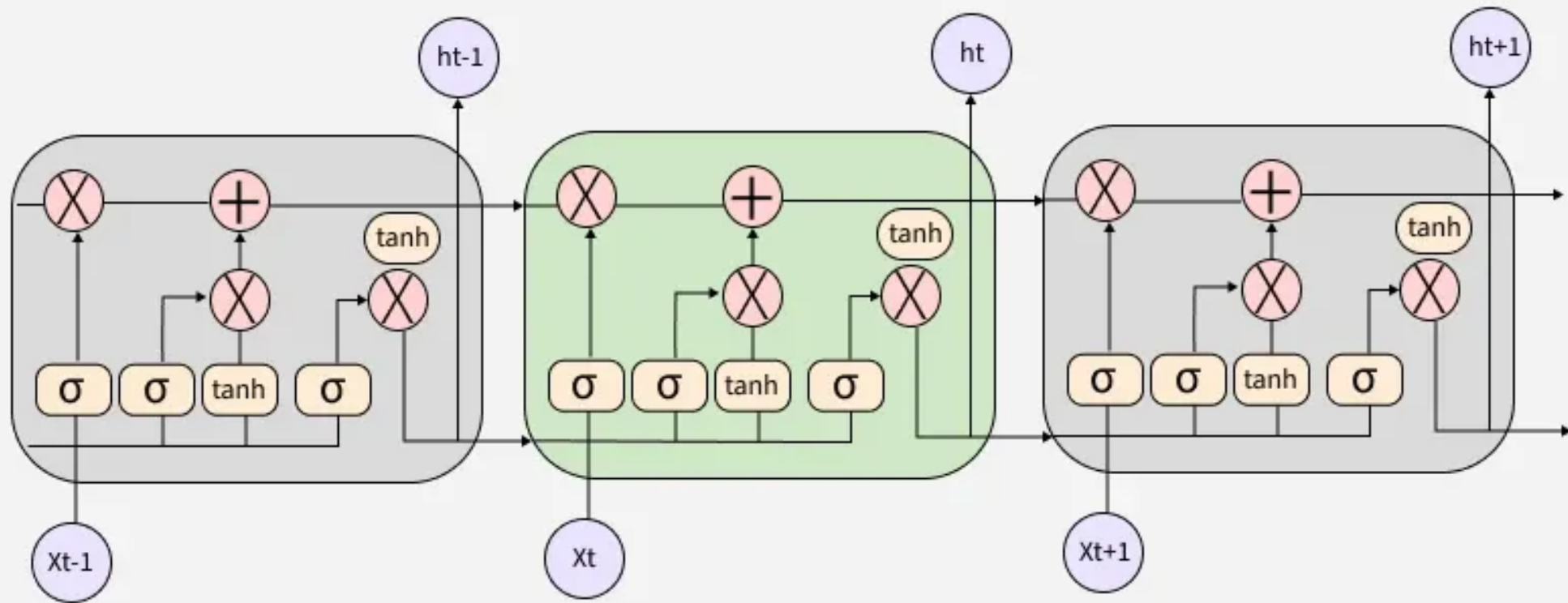
- Lasso: regularization technique
- MAR Lasso (*Jiang, Shen, Li, Gao, 2024*): further reduces # of parameters

Methodology (4): MAR(1) is enough!

- IV, obtained by inverting the Black-Scholes formula, is just price
- The Weak Efficient Market Hypothesis (*Eugene Fama, 1970*): today's stock prices reflect all the data of past prices
- The First Fundamental Theorem of Asset Pricing (*Harrison, Kreps, 1979*): no arbitrage iff EMM exists under which the discounted price of all traded assets are martingales.

Methodology (5): LSTM

Architecture Of LSTMs



Research Design

- **Data collection:**

- Source: Bloomberg
- Time period: Apr 2016 - May 2026

- **Timeline:**

- Month 1: formalize math + acquire data
- Month 2: feature engineer/build/train/fine-tune models
- Month 3: test/evaluate/compare models
- Month 4: document
- Month 5 & 6: validate and revise

Expected Outcomes

- Anticipated findings:
 - LSTM outperforms MAR Lasso
 - MAR Lasso outperforms MAR/VAR
- Potential limitations:
 - "Black box" of deep learning
- Significance:
 - Academic: keep searching for econometric models
 - Practical: tolerate and quantify risks